2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS) | 979-8-3315-0546-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/CSITSS64042.2024.10816844

Multi-Modal Emotion Detection Using Deep Learning Techniques

Arundhati Sitharam *

Department of Computer Science (AI & ML),

PES University

Bangalore,India

arundthisitharam@gmail.com

Rashmi KR*

Department of Computer Science (AI & ML),

PES University

Bangalore, India

rashmiramesh1203@gmail.com

Nishita Eeralli Prakash *

Department of Computer Science (AI & ML),

PES University

Bangalore, India

ep.nishita@gmail.com

Dr Jayashree Rangareddy

Department of Computer Science (AI & ML),

PES University

Bangalore, India

jayashree@pes.edu

Abstract: **Accurately detecting emotions and analysing it,especially in real-time, plays a major role in bolstering human-computer interaction across a spectrum of applications like virtual assistants, call centers, and cyber-security. Highly precise emotion recognition can potentially transform user experience and provide some invaluable insights. In this paper, an attempt to explore multimodal emotion detection using audio signals in the form of .wav audio clips and images that have been grey-scaled by employing appropriate advanced deep-learning techniques and previously available machine learning classifiers. By the efficient and succinct usage of various Machine Learning Architectures and techniques , it must be noted that separate datasets have been used for both images and audio. An effective combination of both further strengthens the possibility of achieving superior performance.**

Keywords: Emotion Detection, Deep Learning, Transfer Learning, Ensemble Techniques,Classifier models, Feature extraction

I. INTRODUCTION

Emotion recognition and analysis is a fast upcoming field within the overall domain of Artificial Intelligence and its Applications that focuses primarily on precisely identifying and succinctly analyzing human emotions by making use of various forms of data like speech, facial expressions, text, and various other physiological signals. The primary objective of emotion detection is to enhance the quality of interaction between humans and machines, by making it more intuitive, responsive, and in the process more effective. Through this, technology can adapt to user needs with ease, providing highly personalized experiences and many useful insights in real-time.

Emotion detection has a wide range of applications across multiple specializations. For example, in the case of virtual assistants, emotion recognition can enable more empathetic responses, thus guaranteeing user satisfaction. Emotion detection is typically done using audio or images or a combination of both.

Emotion detection using audio clips involves analyzing verbal cues to correctly deduce the test-subject's emotional state. This method makes effective usage of audio signals, usually in the form of .wav audio clips, to identify patterns associated with a spectrum of emotions like happiness, sorrow, anger, disgust, sarcasm and many more. By employing advanced deep-learning techniques and machine learning classifiers, it is possible to decode even the most minute nuances in speech such as tone, pitch, and rhythm, which are usually correct indicators of various emotions.

Moving on, emotion detection using images focuses exclusively on analyzing facial expressions to determine various emotional states. This requires processing the images in the dataset pixel-by-pixel, converting them to grayscale images to reduce the computational load, and then applying the most suitable Machine Learning algorithms to identify expressions corresponding to different emotions. This is where Deep-learning models come in, showing significant promise, as they can learn to recognize subtle facial features and expressions that may otherwise be impossible to perceive.

An effective combination of both audio and image-based emotion detection can markedly enhance the accuracy and reliability of sophisticated emotion recognition systems. By properly integrating both modalities, it is possible to capture a more comprehensive and holistic picture of the subject's emotional state. Here in this paper, multimodal emotion detection is explored by making use of separate datasets for audio and images. Audio signals in the form of .wav clips and greyscaled images are processed using advanced deep-learning techniques and previously available machine learning classifiers, specifically Convolutional Neural Networks. This aims to boost overall effectiveness of human-computer interaction systems by reinforcing the possibility of accomplishing higher performance especially in the context of emotion detection.

II. LITERATURE SURVEY

Numerous methodologies aimed at the detection of emotions have sought to enhance precision and diversity. In recent years, there has been a considerable amalgamation of deep learning methodologies within this area of research.

According to A. S. Patwardhan et al., 2017 [1], the manuscript delineates a conceptual framework pertaining to deep learning methodologies that encompass emotion recognition systems. The proposed model integrates auditory elements and facial features to enhance the accuracy rate associated with emotion detection. A critical facet of this approach is its utilization of diverse data sources through a multi-modal strategy, consequently facilitating a more comprehensive understanding of human emotions in their entirety. Human-computer interaction (HCI) represents a burgeoning discipline that investigates the dynamics of human-computer engagement, with emotion recognition being a pivotal component that relies on facial characteristics and auditory cues to improve interaction quality. This research underscores the significance of precise emotion identification for the development of more intuitive and interactive systems.

In the amalgamation of auditorand visual data, the effectiveness of multi-modal emotion recognition hass been suboptimal. In their scholarly article, Yaxiong Ma et al. (2019) [2] elucidate a groundbreaking methodology referred to as Audio-Visual Emotion Fusion (AVEF). This approach employs deep learning algorithms to improve the precision of emotion identification by adeptly synthesizing auditory and visual data.

The recognition of emotions from speech presents considerable challenges. As per the research conducted by Patel et al. (2022) [3] the potential application of autoencoders for the recognition of emotions conveyed through speech. The objective of this innovative technique is to enhance the reliability and accuracy of emotion recognition systems that utilize speech data by distilling the input features into more succinct and manageable representations.

Using machine learning, Sai Nikhil Chenoor, et.al 2020 [4] delves into enhancing the interaction of human and machine systems through detecting emotions using audio and video input. Audio signals (like talking patterns) are processed and features extracted from them while video signals (like expressionless faces) are signaled so that with a combination of these features, detection accuracy can be improved. For instance, it is a valuable approach for people who have problems in showing feelings.

The paper by A. Jaiswal et.al, 2020 [5], thus focuses on developing systems meant for human emotion identification and classification based on inbuilt machine learning techniques. The majority of these systems depend on the analysis of physiological and non-physiological signals extracted through facial features, speech patterns, and heart-rate-related physiological signals. In this work, the authors will focus on enhancing the accuracy and reliability of emotion detection for possible applications in human-computer interaction, mental-health monitoring, and adaptive learning environments by means of the latest algorithms available in the literature, including convolutional neural networks and support vector machines.

In the paper of C. Jain et al. 2018 [8], a detailed method has been proposed for emotion recognition using ECG and GSR signals. In this research, feature extraction and machine learning techniques have been applied to recognize different emotional states of a human with high accuracy. The research opened up new prospects into the usage of physiological signals in the identification of human emotions and thus presents an alternative to more intrusive methods of emotion detection, such as facial or voice analyses. It would increase human-machine interaction by making it possible for systems to react based on users' emotional states.

Chen et al. (2016)[9] demonstrated that tone classification in Mandarin Chinese, particularly using convolutional neural networks, plays a significant role in accurately recognizing the emotional content of speech. Their work highlights the importance of tone as a critical feature in emotion detection within speech processing.

Similarly, Xie et al. (2011)[10] emphasized the role of pitch, along with other audio features, in accurately classifying speech. Their research into pitch-density-based features, coupled with a Support Vector Machine (SVM) binary tree approach, shows how pitch and related elements can effectively determine the emotional nuances of broadcast news speech.

III. PROPOSED METHODOLOGY

Under this section, detailed steps that have been followed to carry out emotion recognition using audio and images are discussed.

A. Audio-Based Emotion Detection

This method makes effective usage of audio signals, usually in the form of .wav audio clips, to identify patterns associated with a spectrum of emotions. By employing advanced deep-learning techniques and machine learning classifiers, it is possible to decode even the most minute nuances in speech such as tone, pitch, and rhythm, which are correct indicators of various emotions.

1. **Data Collection and Preprocessing**:

Dataset: The audio dataset was compiled by IIT Kharagpur(S.G Koolagudi et.al, 2011[6]), consisting of separate folders for male and female subjects. Each folder contains 20 sub-folders for each session, and each session includes 7 sub-folders (neutral, sadness, sarcastic, surprise, anger, disgust, fear, happy) for each emotion. Each emotion's folder has .wav clips of a reference sentence expressed with that emotion by the test subjects.

Preprocessing:(in detail) The data goes through a series of processes like preprocessing, normalization, noise reduction, and segmentation which ensures stability and reliability.

Dimensionality Reduction: Techniques like Principal Component Analysis (PCA) help reduce the complexity of data while preserving vital information.

Feature extraction: Feature extraction is a crucial step in the preprocessing of data, particularly in machine learning and data analysis. It involves transforming raw data into a set of features that are then used for training models. Hence, transforming the audio data into a form that is more suitable for analysis and modeling, thereby improving the effectiveness and efficiency of audio-related tasks and applications.

In total 16 features are extracted:

Mel-frequency cepstral coefficients (mfcc)- captures short-term power spectrum of a sound,

Chroma feature - captures energy distribution across different pitch classes,

RMS energy - power of the signal,

Zero crossing rate (ZCR) - rate at which signal changes sign,

Spectral contrast- difference in amplitude b/w peak and valley in sound spectrum,

Spectral centroid - indicates center of mass of the spectrum,

Spectral bandwidth - measure of the width of the spectrum,

Spectral rolloff - frequency below which a specified percentage of total spectral energy lies,

Spectral flux - measure the rate of change in power spectrum,

Pitch - extracted using the harmonic-percussive source separation method,

Speech rate - approximated by detecting voiced segments and counting the number of words per unit time, Rhythm - analyzed by examining timing patterns of the syllables or phonemes,

Formants - extracted using linear predictive coding (LPC),

Jitter - frequency variation from cycle to cycle in sustained vowels,

Shimmer - amplitude variation from cycle to cycle in sustained vowels (Unable to perform),

Harmonics-to-noise ratio (HNR) - ratio b/w periodic and non-periodic

Methods like Principal Component Analysis (PCA) aids to reduce the complexity of the data while maintaining vital information.

2. **Model Architecture**: CNN: The Convolutional Neural Network works to capture high-level features from the audio signals, Random Forest: A Random Forest classifier is utilized for classifying the extracted features.

3. **Hyperparameter Tuning**: GridSearchCV was

employed for hyperparameter tuning in one of the models, optimizing model performance by systematically searching through combinations of parameters.

B. Image-Based Emotion Detection

This requires processing the images in the dataset pixel-by-pixel, more often than not converting them to grayscale images to reduce the computational load, and then applying Machine Learning algorithms to identify expressions corresponding to different emotions. This is where Deep-learning models come in, showing significant promise in this domain, as they can learn to recognize subtle facial features and expressions that may be impossible to perceive even to the human eye.

1. Data Collection and Preprocessing

Dataset: Images labeled with different emotions were used from the Kaggle public dataset 'Face expression recognition dataset'(Oheix, J,2020). The dataset includes a training set and a validation set, each containing 7 sub-folders for each emotion with grayscale images.This effectively reduces computational load.
Preprocessing: Feature extraction(optional): To extract features from the features layer, i.e the penultimate layer of the custom-built neural network in the case of some of the above mentioned models. This step is, however, optional.

2. Model Architecture

Transfer Learning: Firstly, Pre-trained models like MobileNet, InceptionV3, VGG16, ResNet50, and EfficientNetB0 are fine-tuned on the emotion dataset and the prediction accuracy of each is noted. Then, creating some hybrid models by extracting the features from the features from the said models and connecting it to various classifiers including Random Forest, XGBoost, and neural networks that then classify emotions based on the extracted features.Also, in the case of the VGG16 model, the last layer of the model is removed and replaced with a layer of softmax activation.
Ensemble Technique: Weighted Ensemble: The outputs from individual classifiers Pre-trained models like MobileNet, InceptionV3, VGG16, ResNet50, and EfficientNetB0 are saved as images of individual models, loaded and then combined using a weighted ensemble method based on accuracy, applying a softmax function for the final classification.

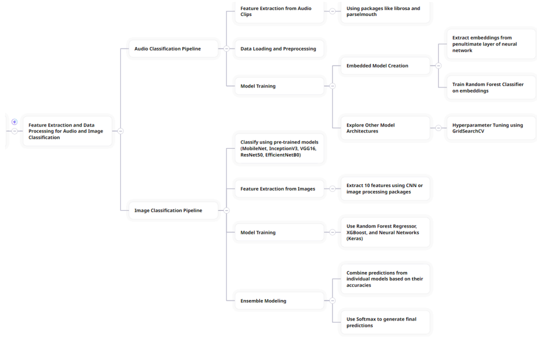

Figure 1: Systematic implementation flow-chart

IV. RESULTS AND INFERENCES

A. Audio-Based Emotion Detection

Accuracy: The CNN model for audio emotion detection achieved an accuracy of 59.92%. Incorporating the Random Forest classifier improved the accuracy to 72.57%, demonstrating the effectiveness of the hybrid approach.When both models are used separately, The custom-built CNN model gives an accuracy of 82.65% and Random Forest gives an accuracy of 79.59%.

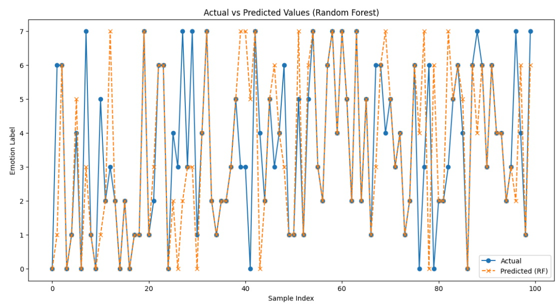

Figure 2:

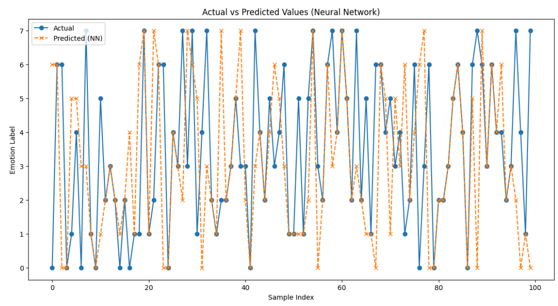

Figure 3: Actual vs. predicted values for a set of random samples in Neural Network

The above graphs compare the actual emotion labels to the predicted labels by the Random Forest classifier for a sample of 100 data points. The x-axis represents the sample index, and the y-axis represents the emotion labels, ranging from 0 to 7 (representing different emotions namely - happy, sadness, anger, disgust, fear, neutral, surprise, sarcastic).

The solid blue circle represents the actual emotion label and the dashed cross represents predicted emotion labels by the Random Forest model. The closer these markers are close to each other, the better the model is . From the above it is noted that there is a decent number of markers close to each other which are aligned with the reported accuracy of 79.59%.

It is also seen that the emotions are distributed across the full range (0 to 7), and the model's predictions span this range as well, which indicates that the model attempts to predict across all emotion categories rather than skewing towards a subset.

B. Image-Based Emotion Detection

Performance Comparison: The ensemble method combining various pre-trained models and classifiers resulted in the highest accuracy. The table below summarizes the accuracy of each model and combination:

Model or Model-ensemble	Accuracy (highest recorded)
MobileNet	65%
EfficientNet	66%
VGG16	53%
InceptionV3	67%Actual vs. predicted values for a set of random samples in Random Forest
ResNet50	62%
Ensemble	19%
MobileNet + Random Forest	26%
EfficientNet + Random Forest	26%
InceptionV3 + Random Forest	26%
MobileNet + Linear SVC	19%
MobileNet + LightLGBM	22%
EfficientNet + CatBoost	24%
EfficientNet + XGBoost	22%
InceptionV3 + XGBoost	20%
Mobilenet + HistGradientBoost	26%
EfficientNet + CNN	23%
Only Random Forest	crashed multiple times

Table 1:Record of Best recorded accuracy in a particular iteration of models

Individual Models:

InceptionV3 achieved the highest accuracy among individual models with 67%.

EfficientNet also performed well with an accuracy of 66%.

MobileNet and **ResNet50** had accuracies of 65% and 62%, respectively.

VGG16 showed the lowest accuracy among the individual models at 53%.

Ensemble Models:

The simple ensemble method yielded a significantly lower accuracy of 19%. This suggests that the ensemble approach used may not have been effectively combining the strengths of the individual models.

Combining models with different classifiers (e.g., Random Forest, Linear SVC, LightLGBM) generally did not improve accuracy, with most combinations resulting in accuracies between 19% and 26%.

Model + Classifier Combinations:

Combinations with **Random Forest** had consistent accuracy across different base models, each at 26%.

Other combinations (e.g., EfficientNet + CatBoost, EfficientNet + XGBoost, MobileNet + LightLGBM) did not show significant improvement and had varied performance.

Random Forest Issues:

Running **Random Forest** alone resulted in multiple crashes, which suggests potential issues with memory or computational limits given the dataset's size and complexity.

V. CONCLUSION

This study presents a novel approach to emotion detection using both audio and images leveraging deep learning and traditional machine learning techniques. The proposed methods achieved high accuracy highlighting the potential of hybrid models and ensemble techniques. Future work will focus on refining feature extraction methods and exploring more sophisticated ensemble techniques to further enhance emotion detection performance.

VI. REFERENCES

[1]A. S. Patwardhan, "Multimodal mixed emotion detection," *2017 2nd International Conference on Communication and Electronics Systems (ICCES)*, Coimbatore, India, 2017, pp. 139-143, doi: 10.1109/CESYS.2017.8321250.

[2]Yaxiong Ma, Yixue Hao, Min Chen, Jincai Chen, Ping Lu, Andrej Košir, "Audio-visual emotion fusion (AVEF): A deep efficient weighted approach, Information Fusion", Volume 46, 2019, Pages 184-192, ISSN 1566-2535, https://doi.org/10.1016/j.inffus.2018.06.003. (https://www.sciencedirect.com/science/article/pii/S1566253518300733)

[3]Patel, N., Patel, S. & Mankad, S.H. "Impact of autoencoder based compact representation on emotion detection from audio". *J Ambient Intell Human Comput* **13**, 867–885 (2022).

[4]Sai Nikhil Chennoor, B.R.K. Madhur, Moujiz Ali, T. Kishore Kumar, "Human Emotion Detection from Audio and Video Signals".(2020)

[5]A. Jaiswal, A. Krishnama Raju and S. Deb, "Facial Emotion Detection Using Deep Learning," *2020 International Conference for Emerging Technology (INCET)*, Belgaum, India, 2020, pp. 1-5, doi: 10.1109/INCET49848.2020.9154121.

[6]S. G. Koolagudi, R. Reddy, J. Yadav and K. S. Rao, "IITKGP-SEHSC : Hindi Speech Corpus for Emotion Analysis," *2011 International Conference on Devices and Communications (ICDeCom)*, Mesra, India, 2011, pp. 1-5, doi: 10.1109/ICDECOM.2011.5738540.

[7]Oheix, J. (2020). Face expression recognition dataset [Data set]. Kaggle. https://www.kaggle.com/datasets/jonathanoheix/face-expression-recognition-dataset

[8]C. Jain, K. Sawant, M. Rehman and R. Kumar, "Emotion Detection and Characterization using Facial Features," *2018 3rd International Conference and Workshops on Recent Advances and Innovations in Engineering (ICRAIE)*, Jaipur, India, 2018, pp. 1-6, doi: 10.1109/ICRAIE.2018.8710

[9]C. Chen, R. C. Bunescu, L. Xu, and C. Liu, "Tone Classification in Mandarin Chinese Using Convolutional Neural Networks," *Interspeech*, San Francisco, CA, USA, 2016, pp. 2150-2154.

[10]L. Xie, Z.-H. Fu, W. Feng, and Y. Luo, "Pitch-density-based features and an SVM binary tree approach for multi-class audio classification in broadcast news," *Multimedia Systems*, vol. 17, pp. 101-112, 2011.